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VACUUM ADIABATIC BODY AND METHOD FOR MANUFACTURING THE REFRIGERATOR

Abstract

A vacuum adiabatic body according to an embodiment may include a first plate, a second plate, and a seal that seals a gap between the first plate and the second plate. Optionally, the vacuum adiabatic body according to an embodiment may include a support that maintains a vacuum space. Optionally, the vacuum adiabatic body according to an embodiment may include a heat transfer resistor that reduces an amount of heat transfer between the first plate and the second plate. The vacuum adiabatic body may include a component coupling portion connected to at least one of the first or second plate so that a component is coupled thereto. Accordingly, the vacuum adiabatic body capable of achieving the industrial purpose may be provided.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation Application of prior U.S. patent application Ser. No. 18/034,874, filed May 1, 2023, which is a U.S. National Stage Application under 35 U.S.C. § 371 of PCT Application No. PCT/KR2021/015503, filed Nov. 1, 2021, which claims priority 35 U.S.C. 119 (a) to Korean Patent Application No. 10-2020-0144745, filed Nov. 2, 2020, whose entire disclosures are hereby incorporated by reference.

BACKGROUND

1. Field

[0002] The present disclosure relates to a vacuum adiabatic body and a method for manufacturing the same.

2. Background

[0003] A vacuum adiabatic wall may be provided to improve adiabatic performance. A device of which at least a portion of an internal space is provided in a vacuum state to achieve an adiabatic effect is referred to as a vacuum adiabatic body.

[0004] The applicant has developed a technology to obtain a vacuum adiabatic body that is capable of being used in various devices and home appliances and has disclosed Korean Application Nos. 10-2015-0109724 and 10-2015-0109722 that relate to the vacuum adiabatic body.

[0005] In the cited document, a plurality of members are coupled to provide a vacuum space. Specifically, a first plate, a conductive resistance sheet, a side plate, and a second plate are sealed to each other. To seal the coupling portion of each member, a sealing process is performed. A small process error occurring in the sealing process leads to vacuum breakage.

[0006] The cited document does not disclose a specific method for insulating a peripheral portion of the vacuum adiabatic body. In particular, a method for manufacturing the vacuum adiabatic body is not described.

BRIEF SUMMARY OF THE INVENTION

[0007] Embodiments provide a vacuum adiabatic body capable of solving limitation of sealing defects by reducing the number of portions to be sealed on a wall of a vacuum space.

[0008] Embodiments also provide a vacuum adiabatic body having high productivity.

[0009] A vacuum adiabatic body according to an embodiment may include a first plate, a second plate, and a seal that seals a gap between the first plate and the second plate. Optionally, the vacuum adiabatic body according to an embodiment may include a support that maintains a vacuum space. Optionally, the vacuum adiabatic body according to an embodiment may include a heat transfer resistor that reduces an amount of heat transfer between the first plate and the second plate. Optionally, the vacuum adiabatic body may include at least one of component coupling portions connected to at least one of the first or second plate so that a component is coupled thereto. Accordingly, the vacuum adiabatic body capable of achieving the industrial purpose may be provided.

[0010] Optionally, a first straight portion and a second straight portion below the first straight portion may be provided in the height direction (y-axis) of the vacuum space. Optionally, a third straight portion may be provided between the first and second straight portions. Optionally, a first

curved portion may be provided between the first and third straight portions. Optionally, a second curved portion may be provided between the third and second straight portions. With such a configuration, strength of the vacuum adiabatic body may increase.

[0011] Optionally, at least a portion of the third straight portion may be in contact with the support. Accordingly, the structural strength of the vacuum adiabatic body may be further strengthened.

[0012] Optionally, the method for manufacturing the vacuum adiabatic body may include a vacuum adiabatic body component preparation process in which components constituting the vacuum adiabatic body are prepared in advance. Optionally, the method for manufacturing the vacuum adiabatic body may include a vacuum adiabatic body component assembly process in which the prepared components are assembled. Optionally, it may be manufactured by a vacuum adiabatic body vacuum exhaust process in which a gas of the vacuum space is discharged after the component assembly process. Optionally, after performing the vacuum adiabatic body component preparation process, at least a portion of the second plate may be in contact with the support. Accordingly, it is possible to increase in structural strength of the vacuum adiabatic body after assembly.

[0013] Optionally, the radiation resistance sheet may not be in contact with the plate. Optionally, the radiation resistance sheet may be in contact with the support. Accordingly, the adiabatic loss may be reduced.

[0014] According to the embodiment, the second plate and the side plate may be processed into the single plate material. Accordingly, the number of sealing positions for coupling the plate may be reduced, and the fear of the vacuum breakage may be largely eliminated.

[0015] According to the embodiment, it is possible to prevent wastage of component, the re-welding, and the reduction in product yield.

[0016] According to the embodiment, the productivity of the vacuum adiabatic body may be improved due to reducing the number of sealing positions, the standardization of the component, the integration of the component, or the exhaust of the plurality of vacuum adiabatic bodies.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] The embodiments will be described in detail with reference to the following drawings in which like reference numerals refer to like elements wherein:

[0018] FIG. 1 is a perspective view of a refrigerator according to an embodiment;

[0019] FIG. 2 is a view schematically illustrating a vacuum adiabatic body used in a main body and a door of the refrigerator;

[0020] FIG. 3 is a view illustrating an example of a support that maintains a vacuum space;

[0021] FIG. 4 is a view for explaining an example of the vacuum with respect to a heat transfer resistor;

[0022] FIG. 5 is a graph illustrating results obtained by observing a process of exhausting the inside of the vacuum adiabatic body with a time and pressure when the support is used;

[0023] FIG. 6 is a graph illustrating results obtained by comparing a vacuum pressure to gas conductivity;

[0024] FIG. 7 is a view illustrating various examples of the vacuum space;

[0025] FIG. 8 is a view for explaining another adiabatic body;

[0026] FIG. 9 is a view for explaining a heat transfer path between first and second plates having different temperatures;

[0027] FIG. 10 is a view for explaining a branch portion on the heat transfer path between first and second plates having different temperatures;

[0028] FIG. 11 is a view for explaining a method for manufacturing a vacuum adiabatic body;

[0029] FIG. **12** is a view for explaining a conductive resistance sheet placed on a heat transfer path;
[0030] FIG. **13** is a view for explaining deformation and displacement of a third straight portion;
[0031] FIG. **14** is a view for explaining a peripheral portion of a radiation resistance sheet;
[0032] FIG. **15** is a side view showing an angle between the first plate and the side plate;
[0033] FIG. **16** is a view illustrating a state after a component assembly process of the vacuum adiabatic body is performed; and
[0034] FIG. **17** is a side view showing an angle between the first plate and the side plate after performing the component assembly process.

DETAILED DESCRIPTION

[0035] Hereinafter, specific embodiments will be described in detail with reference to the accompanying drawings. The invention may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein, and a person of ordinary skill in the art, who understands the spirit of the present invention, may readily implement other embodiments included within the scope of the same concept by adding, changing, deleting, and adding components; rather, it will be understood that they are also included within the scope of the present invention. The present invention may have many embodiments in which the idea is implemented, and in each embodiment, any portion may be replaced with a corresponding portion or a portion having a related action according to another embodiment. The present invention may be any one of the examples presented below or a combination of two or more examples.

[0036] The present disclosure relates to a vacuum adiabatic body including a first plate; a second plate; a vacuum space defined between the first and second plates; and a seal providing the vacuum space that is in a vacuum state. The vacuum space may be a space in a vacuum state provided in an internal space between the first plate and the second plate. The seal may seal the first plate and the second plate to provide the internal space provided in the vacuum state. The vacuum adiabatic body may optionally include a side plate connecting the first plate to the second plate. In the present disclosure, the expression “plate” may mean at least one of the first and second plates or the side plate. At least a portion of the first and second plates and the side plate may be integrally provided, or at least portions may be sealed to each other. Optionally, the vacuum adiabatic body may include a support that maintains the vacuum space. The vacuum adiabatic body may selectively include a thermal insulator that reduces an amount of heat transfer between a first space provided in vicinity of the first plate and a second space provided in vicinity of the second plate or reduces an amount of heat transfer between the first plate and the second plate. Optionally, the vacuum adiabatic body may include a component coupling portion provided on at least a portion of the plate. Optionally, the vacuum adiabatic body may include another adiabatic body. Another adiabatic body may be provided to be connected to the vacuum adiabatic body. Another adiabatic body may be an adiabatic body having a degree of vacuum, which is equal to or different from a degree of vacuum of the vacuum adiabatic body. Another adiabatic body may be an adiabatic body that does not include a degree of vacuum less than that of the vacuum adiabatic body or a portion that is in a vacuum state therein. In this case, it may be advantageous to connect another object to another adiabatic body.

[0037] In the present disclosure, a direction along a wall defining the vacuum space may include a longitudinal direction of the vacuum space and a height direction of the vacuum space. The height direction of the vacuum space may be defined as any one direction among virtual lines connecting the first space to the second space to be described later while passing through the vacuum space. The longitudinal direction of the vacuum space may be defined as a direction perpendicular to the set height direction of the vacuum space. In the present disclosure, that an object A is connected to an object B means that at least a portion of the object A and at least a portion of the object B are directly connected to each other, or that at least a portion of the object A and at least a portion of the object B are connected to each other through an intermedium interposed between the objects A and B. The intermedium may be provided on at least one of the object A or the object B. The

connection may include that the object A is connected to the intermedium, and the intermedium is connected to the object B. A portion of the intermedium may include a portion connected to either one of the object A and the object B. The other portion of the intermedium may include a portion connected to the other of the object A and the object B. As a modified example, the connection of the object A to the object B may include that the object A and the object B are integrally prepared in a shape connected in the above-described manner. In the present disclosure, an embodiment of the connection may be support, combine, or a seal, which will be described later. In the present disclosure, that the object A is supported by the object B means that the object A is restricted in movement by the object B in one or more of the +X, -X, +Y, -Y, +Z, and -Z axis directions. In the present invention, an embodiment of the support may be the combine or seal, which will be described later. In the present invention, that the object A is combined with the object B may define that the object A is restricted in movement by the object B in one or more of the X, Y, and Z-axis directions. In the present disclosure, an embodiment of the combining may be the sealing to be described later. In the present disclosure, that the object A is sealed to the object B may define a state in which movement of a fluid is not allowed at the portion at which the object A and the object B are connected. In the present disclosure, one or more objects, i.e., at least a portion of the object A and the object B, may be defined as including a portion of the object A, the whole of the object A, a portion of the object B, the whole of the object B, a portion of the object A and a portion of the object B, a portion of the object A and the whole of the object B, the whole of the object A and a portion of the object B, and the whole of the object A and the whole of the object B. In the present disclosure, that the plate A may be a wall defining the space A may be defined as that at least a portion of the plate A may be a wall defining at least a portion of the space A. That is, at least a portion of the plate A may be a wall forming the space A, or the plate A may be a wall forming at least a portion of the space A. In the present disclosure, a central portion of the object may be defined as a central portion among three divided portions when the object is divided into three sections based on the longitudinal direction of the object. A periphery of the object may be defined as a portion disposed at a left or right side of the central portion among the three divided portions. The periphery of the object may include a surface that is in contact with the central portion and a surface opposite thereto. The opposite side may be defined as a border or edge of the object. Examples of the object may include a vacuum adiabatic body, a plate, a heat transfer resistor, a support, a vacuum space, and various components to be introduced in the present disclosure. In the present disclosure, a degree of heat transfer resistance may indicate a degree to which an object resists heat transfer and may be defined as a value determined by a shape including a thickness of the object, a material of the object, and a processing method of the object. The degree of the heat transfer resistance may be defined as the sum of a degree of conduction resistance, a degree of radiation resistance, and a degree of convection resistance. The vacuum adiabatic body according to the present disclosure may include a heat transfer path defined between spaces having different temperatures, or a heat transfer path defined between plates having different temperatures. For example, the vacuum adiabatic body according to the present disclosure may include a heat transfer path through which cold is transferred from a low-temperature plate to a high-temperature plate. In the present disclosure, when a curved portion includes a first portion extending in a first direction and a second portion extending in a second direction different from the first direction, the curved portion may be defined as a portion that connects the first portion to the second portion (including 90 degrees).

[0038] In the present disclosure, the vacuum adiabatic body may optionally include a component coupling portion. The component coupling portion may be defined as a portion provided on the plate to which components are connected to each other. The component connected to the plate may be defined as a penetration portion disposed to pass through at least a portion of the plate and a surface component disposed to be connected to a surface of at least a portion of the plate. At least one of the penetration component or the surface component may be connected to the component

coupling portion. The penetration component may be a component that defines a path through which a fluid (electricity, refrigerant, water, air, etc.) passes mainly. In the present disclosure, the fluid is defined as any kind of flowing material. The fluid includes moving solids, liquids, gases, and electricity. For example, the component may be a component that defines a path through which a refrigerant for heat exchange passes, such as a suction line heat exchanger (SLHX) or a refrigerant tube. The component may be an electric wire that supplies electricity to an apparatus. As another example, the component may be a component that defines a path through which air passes, such as a cold duct, a hot air duct, and an exhaust port. As another example, the component may be a path through which a fluid such as coolant, hot water, ice, and defrost water pass. The surface component may include at least one of a peripheral adiabatic body, a side panel, injected foam, a pre-prepared resin, a hinge, a latch, a basket, a drawer, a shelf, a light, a sensor, an evaporator, a front decor, a hotline, a heater, an exterior cover, or another adiabatic body.

[0039] As an example to which the vacuum adiabatic body is applied, the present disclosure may include an apparatus having the vacuum adiabatic body. Examples of the apparatus may include an appliance. Examples of the appliance may include home appliances including a refrigerator, a cooking appliance, a washing machine, a dishwasher, and an air conditioner, etc. As an example in which the vacuum adiabatic body is applied to the apparatus, the vacuum adiabatic body may constitute at least a portion of a body and a door of the apparatus. As an example of the door, the vacuum adiabatic body may constitute at least a portion of a general door and a door-in-door (DID) that is in direct contact with the body. Here, the door-in-door may mean a small door placed inside the general door. As another example to which the vacuum adiabatic body is applied, the present disclosure may include a wall having the vacuum adiabatic body. Examples of the wall may include a wall of a building, which includes a window.

[0040] Hereinafter, the present disclosure will be described in detail with reference to the accompanying drawings. Each of the drawings accompanying the embodiment may be different from, exaggerated, or simply indicated from an actual article, and detailed components may be indicated with simplified features. The embodiment should not be interpreted as being limited only to the size, structure, and shape presented in the drawings. In the embodiments accompanying each of the drawings, unless the descriptions conflict with each other, some configurations in the drawings of one embodiment may be applied to some configurations of the drawings in another embodiment, and some structures in one embodiment may be applied to some structures in another embodiment. In the description of the drawings for the embodiment, the same reference numerals may be assigned to different drawings as reference numerals of specific components constituting the embodiment. Components having the same reference number may perform the same function. For example, the first plate constituting the vacuum adiabatic body has a portion corresponding to the first space throughout all embodiments and is indicated by reference number **10**. The first plate may have the same number for all embodiments and may have a portion corresponding to the first space, but the shape of the first plate may be different in each embodiment. Not only the first plate, but also the side plate, the second plate, and another adiabatic body may be understood as well.

[0041] FIG. **1** is a perspective view of a refrigerator according to an embodiment, and FIG. **2** is a schematic view illustrating a vacuum adiabatic body used for a body and a door of the refrigerator. Referring to FIG. **1**, the refrigerator **1** includes a main body **2** provided with a cavity **9** capable of storing storage goods and a door **3** provided to open and close the main body **2**. The door **3** may be rotatably or slidably disposed to open or close the cavity **9**. The cavity **9** may provide at least one of a refrigerating compartment and a freezing compartment. A cold source that supplies cold to the cavity may be provided. For example, the cold source may be an evaporator **7** that evaporates the refrigerant to take heat. The evaporator **7** may be connected to a compressor **4** that compresses the refrigerant evaporated to the cold source. The evaporator **7** may be connected to a condenser **5** that condenses the compressed refrigerant to the cold source. The evaporator **7** may be connected to an expander **6** that expands the refrigerant condensed in the cold source. A fan corresponding to the

evaporator and the condenser may be provided to promote heat exchange. As another example, the cold source may be a heat absorption surface of a thermoelectric element. A heat absorption sink may be connected to the heat absorption surface of the thermoelectric element. A heat sink may be connected to a heat radiation surface of the thermoelectric element. A fan corresponding to the heat absorption surface and the heat generation surface may be provided to promote heat exchange.

[0042] Referring to FIG. 2, plates **10**, **15**, and **20** may be walls defining the vacuum space. The plates may be walls that partition the vacuum space from an external space of the vacuum space. An example of the plates is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples.

[0043] The plate may be provided as one portion or may be provided to include at least two portions connected to each other. As a first example, the plate may include at least two portions connected to each other in a direction along a wall defining the vacuum space. Any one of the two portions may include a portion (e.g., a first portion) defining the vacuum space. The first portion may be a single portion or may include at least two portions that are sealed to each other. The other one of the two portions may include a portion (e.g., a second portion) extending from the first portion of the first plate in a direction away from the vacuum space or extending in an inner direction of the vacuum space. As a second example, the plate may include at least two layers connected to each other in a thickness direction of the plate. Any one of the two layers may include a layer (e.g., the first portion) defining the vacuum space. The other one of the two layers may include a portion (e.g., the second portion) provided in an external space (e.g., a first space and a second space) of the vacuum space. In this case, the second portion may be defined as an outer cover of the plate. The other one of the two layers may include a portion (e.g., the second portion) provided in the vacuum space. In this case, the second portion may be defined as an inner cover of the plate.

[0044] The plate may include a first plate **10** and a second plate **20**. One surface of the first plate (the inner surface of the first plate) provides a wall defining the vacuum space, and the other surface (the outer surface of the first plate) of the first plate A wall defining the first space may be provided. The first space may be a space provided in the vicinity of the first plate, a space defined by the apparatus, or an internal space of the apparatus. In this case, the first plate may be referred to as an inner case. When the first plate and the additional member define the internal space, the first plate and the additional member may be referred to as an inner case. The inner case may include two or more layers. In this case, one of the plurality of layers may be referred to as an inner panel. One surface of the second plate (the inner surface of the second plate) provides a wall defining the vacuum space, and the other surface (the outer surface of the first plate) of the second plate A wall defining the second space may be provided. The second space may be a space provided in vicinity of the second plate, another space defined by the apparatus, or an external space of the apparatus. In this case, the second plate may be referred to as an outer case. When the second plate and the additional member define the external space, the second plate and the additional member may be referred to as an outer case. The outer case may include two or more layers. In this case, one of the plurality of layers may be referred to as an outer panel. The second space may be a space having a temperature higher than that of the first space or a space having a temperature lower than that of the first space. Optionally, the plate may include a side plate **15**. In FIG. 2, the side plate may also perform a function of a conductive resistance sheet **60** to be described later, according to the disposition of the side plate. The side plate may include a portion extending in a height direction of a space defined between the first plate and the second plate or a portion extending in a height direction of the vacuum space. One surface of the side plate may provide a wall defining the vacuum space, and the other surface of the side plate may provide a wall defining an external space of the vacuum space. The external space of the vacuum space may be at least one of the first space or the second space or a space in which another adiabatic body to be described later is disposed. The side plate may be integrally provided by extending at least one of the first plate or the second

plate or a separate component connected to at least one of the first plate or the second plate.

[0045] The plate may optionally include a curved portion. In the present disclosure, the plate including a curved portion may be referred to as a bent plate. The curved portion may include at least one of the first plate, the second plate, the side plate, between the first plate and the second plate, between the first plate and the side plate, or between the second plate and the side plate. The plate may include at least one of a first curved portion or a second curved portion, an example of which is as follows. First, the side plate may include the first curved portion. A portion of the first curved portion may include a portion connected to the first plate. Another portion of the first curved portion may include a portion connected to the second curved portion. In this case, a curvature radius of each of the first curved portion and the second curved portion may be large. The other portion of the first curved portion may be connected to an additional straight portion or an additional curved portion, which are provided between the first curved portion and the second curved portion. In this case, a curvature radius of each of the first curved portion and the second curved portion may be small. Second, the side plate may include the second curved portion. A portion of the second curved portion may include a portion connected to the second plate. The other portion of the second curved portion may include a portion connected to the first curved portion. In this case, a curvature radius of each of the first curved portion and the second curved portion may be large. The other portion of the second curved portion may be connected to an additional straight portion or an additional curved portion, which are provided between the first curved portion and the second curved portion. In this case, a curvature radius of each of the first curved portion and the second curved portion may be small. Here, the straight portion may be defined as a portion having a curvature radius greater than that of the curved portion. The straight portion may be understood as a portion having a perfect plane or a curvature radius greater than that of the curved portion. Third, the first plate may include the first curved portion. A portion of the first curved portion may include a portion connected to the side plate. A portion connected to the side plate may be provided at a position that is away from the second plate at a portion at which the first plate extends in the longitudinal direction of the vacuum space. Fourth, the second plate may include the second curved portion. A portion of the second curved portion may include a portion connected to the side plate. A portion connected to the side plate may be provided at a position that is away from the first plate at a portion at which the second plate extends in the longitudinal direction of the vacuum space. The present disclosure may include a combination of any one of the first and second examples described above and any one of the third and fourth examples described above.

[0046] In the present disclosure, the vacuum space **50** may be defined as a third space. The vacuum space may be a space in which a vacuum pressure is maintained. In the present disclosure, the expression that a vacuum degree of A is higher than that of B means that a vacuum pressure of A is lower than that of B.

[0047] In the present disclosure, the seal **61** may be a portion provided between the first plate and the second plate. Examples of sealing are as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. The sealing may include fusion welding for coupling the plurality of objects by melting at least a portion of the plurality of objects. For example, the first plate and the second plate may be welded by laser welding in a state in which a melting bond such as a filler metal is not interposed therebetween, a portion of the first and second plates and a portion of the component coupling portion may be welded by high-frequency brazing or the like, or a plurality of objects may be welded by a melting bond that generates heat. The sealing may include pressure welding for coupling the plurality of objects by a mechanical pressure applied to at least a portion of the plurality of objects. For example, as a component connected to the component coupling portion, an object made of a material having a degree of deformation resistance less than that of the plate may be pressure-welded by a method such as pinch-off.

[0048] A machine room **8** may be optionally provided outside the vacuum adiabatic body. The machine room may be defined as a space in which components connected to the cold source are accommodated. Optionally, the vacuum adiabatic body may include a port **40**. The port may be provided at any one side of the vacuum adiabatic body to discharge air of the vacuum space **50**. Optionally, the vacuum adiabatic body may include a conduit **64** passing through the vacuum space **50** to install components connected to the first space and the second space.

[0049] FIG. **3** is a view illustrating an example of a support that maintains the vacuum space. An example of the support is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples.

[0050] The supports **30**, **31**, **33**, and **35** may be provided to support at least a portion of the plate and a heat transfer resistor to be described later, thereby reducing deformation of at least some of the vacuum space **50**, the plate, and the heat transfer resistor to be described later due to external force. The external force may include at least one of a vacuum pressure or external force excluding the vacuum pressure. When the deformation occurs in a direction in which a height of the vacuum space is lower, the support may reduce an increase in at least one of radiant heat conduction, gas heat conduction, surface heat conduction, or support heat conduction, which will be described later. The support may be an object provided to maintain a gap between the first plate and the second plate or an object provided to support the heat transfer resistor. The support may have a degree of deformation resistance greater than that of the plate or be provided to a portion having weak degree of deformation resistance among portions constituting the vacuum adiabatic body, the apparatus having the vacuum adiabatic body, and the wall having the vacuum adiabatic body. According to an embodiment, a degree of deformation resistance represents a degree to which an object resists deformation due to external force applied to the object and is a value determined by a shape including a thickness of the object, a material of the object, a processing method of the object, and the like. Examples of the portions having the weak degree of deformation resistance include the vicinity of the curved portion defined by the plate, at least a portion of the curved portion, the vicinity of an opening defined in the body of the apparatus, which is provided by the plate, or at least a portion of the opening. The support may be disposed to surround at least a portion of the curved portion or the opening or may be provided to correspond to the shape of the curved portion or the opening. However, it is not excluded that the support is provided in other portions. The opening may be understood as a portion of the apparatus including the body and the door capable of opening or closing the opening defined in the body.

[0051] An example in which the support is provided to support the plate is as follows. First, at least a portion of the support may be provided in a space defined inside the plate. The plate may include a portion including a plurality of layers, and the support may be provided between the plurality of layers. Optionally, the support may be provided to be connected to at least a portion of the plurality of layers or be provided to support at least a portion of the plurality of layers. Second, at least a portion of the support may be provided to be connected to a surface defined on the outside of the plate. The support may be provided in the vacuum space or an external space of the vacuum space. For example, the plate may include a plurality of layers, and the support may be provided as any one of the plurality of layers. Optionally, the support may be provided to support the other one of the plurality of layers. For example, the plate may include a plurality of portions extending in the longitudinal direction, and the support may be provided as any one of the plurality of portions. Optionally, the support may be provided to support the other one of the plurality of parts. As further another example, the support may be provided in the vacuum space or the external space of the vacuum space as a separate component, which is distinguished from the plate. Optionally, the support may be provided to support at least a portion of a surface defined on the outside of the plate. Optionally, the support may be provided to support one surface of the first plate and one surface of the second plate, and one surface of the first plate and one surface of the second plate may be provided to face each other. Third, the support may be provided to be integrated with the

plate. An example in which the support is provided to support the heat transfer resistor may be understood instead of the example in which the support is provided to support the plate. A duplicated description will be omitted.

[0052] An example of the support in which heat transfer through the support is designed to be reduced is as follows. First, at least a portion of the components disposed in the vicinity of the support may be provided so as not to be in contact with the support or provided in an empty space provided by the support. Examples of the components include a tube or component connected to the heat transfer resistor to be described later, an exhaust port, a getter port, a tube or component passing through the vacuum space, or a tube or component of which at least a portion is disposed in the vacuum space. Examples of the empty space may include an empty space provided in the support, an empty space provided between the plurality of supports, and an empty space provided between the support and a separate component that is distinguished from the support. Optionally, at least a portion of the component may be disposed in a through-hole defined in the support, be disposed between the plurality of bars, be disposed between the plurality of connection plates, or be disposed between the plurality of support plates. Optionally, at least a portion of the component may be disposed in a spaced space between the plurality bars, be disposed in a spaced space between the plurality of connection plates, or be disposed in a spaced space between the plurality of support plates. Second, the adiabatic body may be provided on at least a portion of the support or in the vicinity of at least a portion of the support. The adiabatic body may be provided to be in contact with the support or provided so as not to be in contact with the support. The adiabatic body may be provided at a portion in which the support and the plate are in contact with each other. The adiabatic body may be provided on at least a portion of one surface and the other surface of the support or be provided to cover at least a portion of one surface and the other surface of the support. The adiabatic body may be provided on at least a portion of a periphery of one surface and a periphery of the other surface of the support or be provided to cover at least a portion of a periphery of one surface and a periphery of the other surface of the support. The support may include a plurality of bars, and the adiabatic body may be disposed on an area from a point at which any one of the plurality of bars is disposed to a midpoint between the one bar and the surrounding bars. Third, when cold is transferred through the support, a heat source may be disposed at a position at which the heat adiabatic body described in the second example is disposed. When a temperature of the first space is lower than a temperature of the second space, the heat source may be disposed on the second plate or in the vicinity of the second plate. When heat is transmitted through the support, a cold source may be disposed at a position at which the heat adiabatic body described in the second example is disposed. When a temperature of the first space is higher than a temperature of the second space, the cold source may be disposed on the second plate or in the vicinity of the second plate. As fourth example, the support may include a portion having heat transfer resistance higher than a metal or a portion having heat transfer resistance higher than the plate. The support may include a portion having heat transfer resistance less than that of another adiabatic body. The support may include at least one of a non-metal material, PPS, and glass fiber (GF), low outgassing PC, PPS, or LCP. This is done for a reason in which high compressive strength, low outgassing, and a water absorption rate, low thermal conductivity, high compressive strength at a high temperature, and excellent workability are being capable of obtained.

[0053] Examples of the support may be the bars **30** and **31**, the connection plate **35**, the support plate **35**, a porous material **33**, and a filler **33**. In this embodiment, the support may include any one of the above examples, or an example in which at least two examples are combined. As first example, the support may include bars **30** and **31**. The bar may include a portion extending in a direction in which the first plate and the second plate are connected to each other to support a gap between the first plate and the second plate. The bar may include a portion extending in a height direction of the vacuum space and a portion extending in a direction that is substantially perpendicular to the direction in which the plate extends. The bar may be provided to support only

one of the first plate and the second plate or may be provided both the first plate and the second plate. For example, one surface of the bar may be provided to support a portion of the plate, and the other surface of the bar may be provided so as not to be in contact with the other portion of the plate. As another example, one surface of the bar may be provided to support at least a portion of the plate, and the other surface of the bar may be provided to support the other portion of the plate. The support may include a bar having an empty space therein or a plurality of bars, and an empty space are provided between the plurality of bars. In addition, the support may include a bar, and the bar may be disposed to provide an empty space between the bar and a separate component that is distinguished from the bar. The support may selectively include a connection plate **35** including a portion connected to the bar or a portion connecting the plurality of bars to each other. The connection plate may include a portion extending in the longitudinal direction of the vacuum space or a portion extending in the direction in which the plate extends. An XZ-plane cross-sectional area of the connection plate may be greater than an XZ-plane cross-sectional area of the bar. The connection plate may be provided on at least one of one surface and the other surface of the bar or may be provided between one surface and the other surface of the bar. At least one of one surface and the other surface of the bar may be a surface on which the bar supports the plate. The shape of the connection plate is not limited. The support may include a connection plate having an empty space therein or a plurality of connection plates, and an empty space are provided between the plurality of connection plates. In addition, the support may include a connection plate, and the connection plate may be disposed to provide an empty space between the connection plate and a separate component that is distinguished from the connection plate. As a second example, the support may include a support plate **35**. The support plate may include a portion extending in the longitudinal direction of the vacuum space or a portion extending in the direction in which the plate extends. The support plate may be provided to support only one of the first plate and the second plate or may be provided both the first plate and the second plate. For example, one surface of the support plate may be provided to support a portion of the plate, and the other surface of the support plate may be provided so as not to be in contact with the other portion of the plate. As another example, one surface of the support plate may be provided to support at least a portion of the plate, and the other surface of the support plate may be provided to support the other portion of the plate. A cross-sectional shape of the support plate is not limited. The support may include a support plate having an empty space therein or a plurality of support plates, and an empty space are provided between the plurality of support plates. In addition, the support may include a support plate, and the support plate may be disposed to provide an empty space between the support plate and a separate component that is distinguished from the support plate. As a third example, the support may include a porous material **33** or a filler **33**. The inside of the vacuum space may be supported by the porous material or the filler. The inside of the vacuum space may be completely filled by the porous material or the filler. The support may include a plurality of porous materials or a plurality of fillers, and the plurality of porous materials or the plurality of fillers may be disposed to be in contact with each other. When an empty space is provided inside the porous material, provided between the plurality of porous materials, or provided between the porous material and a separate component that is distinguished from the porous material, the porous material may be understood as including any one of the aforementioned bar, connection plate, and support plate. When an empty space is provided inside the filler, provided between the plurality of fillers, or provided between the filler and a separate component that is distinguished from the filler, the filler may be understood as including any one of the aforementioned bar, connection plate, and support plate. The support according to the present disclosure may include any one of the above examples or an example in which two or more examples are combined.

[0054] Referring to FIG. **3a**, as an embodiment, the support may include a bar **31** and a connection plate and support plate **35**. The connection plate and the supporting plate may be designed separately. Referring to FIG. **3b**, as an embodiment, the support may include a bar **31**, a connection

plate and support plate **35**, and a porous material **33** filled in the vacuum space. The porous material **33** may have emissivity greater than that of stainless steel, which is a material of the plate, but since the vacuum space is filled, resistance efficiency of radiant heat transfer is high. The porous material may also function as a heat transfer resistor to be described later. More preferably, the porous material may perform a function of a radiation resistance sheet to be described later. Referring to FIG. **3c**, as an embodiment, the support may include a porous material **33** or a filler **33**. The porous material **33** and the filler may be provided in a compressed state to maintain a gap between the vacuum space. The film **34** may be provided in a state in which a hole is punched as, for example, a PE material. The porous material **33** or the filler may perform both a function of the heat transfer resistor and a function of the support, which will be described later. More preferably, the porous material may perform both a function of the radiation resistance sheet and a function of the support to be described later.

[0055] FIG. **4** is a view for explaining an example of the vacuum adiabatic body based on heat transfer resistors **32**, **33**, **60**, and **63** (e.g., thermal insulator and a heat transfer resistance body). The vacuum adiabatic body according to the present disclosure may optionally include a heat transfer resistor. An example of the heat transfer resistor is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples.

[0056] The heat transfer resistors **32**, **33**, **60**, and **63** may be objects that reduce an amount of heat transfer between the first space and the second space or objects that reduce an amount of heat transfer between the first plate and the second plate. The heat transfer resistor may be disposed on a heat transfer path defined between the first space and the second space or be disposed on a heat transfer path formed between the first plate and the second plate. The heat transfer resistor may include a portion extending in a direction along a wall defining the vacuum space or a portion extending in a direction in which the plate extends. Optionally, the heat transfer resistor may include a portion extending from the plate in a direction away from the vacuum space. The heat transfer resistor may be provided on at least a portion of the periphery of the first plate or the periphery of the second plate or be provided on at least a portion of an edge of the first plate or an edge of the second plate. The heat transfer resistor may be provided at a portion, in which the through-hole is defined, or provided as a tube connected to the through-hole. A separate tube or a separate component that is distinguished from the tube may be disposed inside the tube. The heat transfer resistor may include a portion having heat transfer resistance greater than that of the plate. In this case, adiabatic performance of the vacuum adiabatic body may be further improved. A shield **62** may be provided on the outside of the heat transfer resistor to be insulated. The inside of the heat transfer resistor may be insulated by the vacuum space. The shield may be provided as a porous material or a filler that is in contact with the inside of the heat transfer resistor. The shield may be an adiabatic structure that is exemplified by a separate gasket placed outside the inside of the heat transfer resistor. The heat transfer resistor may be a wall defining the third space.

[0057] An example in which the heat transfer resistor is connected to the plate may be understood as replacing the support with the heat transfer resistor in an example in which the support is provided to support the plate. A duplicate description will be omitted. The example in which the heat transfer resistor is connected to the support may be understood as replacing the plate with the support in the example in which the heat transfer resistor is connected to the plate. A duplicate description will be omitted. The example of reducing heat transfer via the heat transfer body may be applied as a substitute the example of reducing the heat transfer via the support, and thus, the same explanation will be omitted.

[0058] In the present disclosure, the heat transfer resistor may be one of a radiation resistance sheet **32**, a porous material **33**, a filler **33**, and a conductive resistance sheet. In the present disclosure, the heat transfer resistor may include a combination of at least two of the radiation resistance sheet **32**, the porous material **33**, the filler **33**, and the conductive resistance sheet. As a first example, the heat transfer resistor may include a radiation resistance sheet **32**. The radiation resistance sheet may

include a portion having heat transfer resistance greater than that of the plate, and the heat transfer resistance may be a degree of resistance to heat transfer by radiation. The support may perform a function of the radiation resistance sheet together. A conductive resistance sheet to be described later may perform the function of the radiation resistance sheet together. As a second example, the heat transfer resistor may include conduction resistance sheets **60** and **63**. The conductive resistance sheet may include a portion having heat transfer resistance greater than that of the plate, and the heat transfer resistance may be a degree of resistance to heat transfer by conduction. For example, the conductive resistance sheet may have a thickness less than that of at least a portion of the plate. As another example, the conductive resistance sheet may include one end and the other end, and a length of the conductive resistance sheet may be longer than a straight distance connecting one end of the conductive resistance sheet to the other end of the conductive resistance sheet. As another example, the conductive resistance sheet may include a material having resistance to heat transfer greater than that of the plate by conduction. As another example, the heat transfer resistor may include a portion having a curvature radius less than that of the plate.

[0059] Referring to FIG. **4a**, for example, a conductive resistance sheet may be provided on a side plate connecting the first plate to the second plate. Referring to FIG. **4b**, for example, a conductive resistance sheet **60** may be provided on at least a portion of the first plate and the second plate. A connection frame **70** may be further provided outside the conductive resistance sheet. The connection frame may be a portion from which the first plate or the second plate extends or a portion from which the side plate extends. Optionally, the connection frame **70** may include a portion at which a component for sealing the door and the body and a component disposed outside the vacuum space such as the exhaust port and the getter port, which are required for the exhaust process, are connected to each other. Referring to FIG. **4c**, for example, a conductive resistance sheet may be provided on a side plate connecting the first plate to the second plate. The conductive resistance sheet may be installed in a through-hole passing through the vacuum space. The conduit **64** may be provided separately outside the conductive resistance sheet. The conductive resistance sheet may be provided in a pleated shape. Through this, the heat transfer path may be lengthened, and deformation due to a pressure difference may be prevented. A separate shielding member for insulating the conductive resistance sheet **63** may also be provided. The conductive resistance sheet may include a portion having a degree of deformation resistance less than that of at least one of the plate, the radiation resistance sheet, or the support. The radiation resistance sheet may include a portion having a degree of deformation resistance less than that of at least one of the plate or the support. The plate may include a portion having a degree of deformation resistance less than that of the support. The conductive resistance sheet may include a portion having conductive heat transfer resistance greater than that of at least one of the plate, the radiation resistance sheet, or the support. The radiation resistance sheet may include a portion having radiation heat transfer resistance greater than that of at least one of the plate, the conductive resistance sheet, or the support. The support may include a portion having heat transfer resistance greater than that of the plate. For example, at least one of the plate, the conductive resistance sheet, or the connection frame may include stainless steel material, the radiation resistance sheet may include aluminum, and the support may include a resin material.

[0060] FIG. **5** is a graph for observing a process of exhausting the inside of the vacuum adiabatic body with a time and pressure when the support is used. An example of a vacuum adiabatic body vacuum exhaust process vacuum is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples.

[0061] While the exhaust process is being performed, an outgassing process, which is a process in which a gas of the vacuum space is discharged, or a potential gas remaining in the components of the vacuum adiabatic body is discharged, may be performed. As an example of the outgassing process, the exhaust process may include at least one of heating or drying the vacuum adiabatic body, providing a vacuum pressure to the vacuum adiabatic body, or providing a getter to the

vacuum adiabatic body. In this case, it is possible to promote the vaporization and exhaust of the potential gas remaining in the component provided in the vacuum space. The exhaust process may include a process of cooling the vacuum adiabatic body. The cooling process may be performed after the process of heating or drying the vacuum adiabatic body is performed. The process of heating or drying the vacuum adiabatic body process of providing the vacuum pressure to the vacuum adiabatic body may be performed together. The process of heating or drying the vacuum adiabatic body and the process of providing the getter to the vacuum adiabatic body may be performed together. After the process of heating or drying the vacuum adiabatic body is performed, the process of cooling the vacuum adiabatic body may be performed. The process of providing the vacuum pressure to the vacuum adiabatic body and the process of providing the getter to the vacuum adiabatic body may be performed so as not to overlap each other. For example, after the process of providing the vacuum pressure to the vacuum adiabatic body is performed, the process of providing the getter to the vacuum adiabatic body may be performed. When the vacuum pressure is provided to the vacuum adiabatic body, a pressure of the vacuum space may drop to a certain level and then no longer drop. Here, after stopping the process of providing the vacuum pressure to the vacuum adiabatic body, the getter may be input. As an example of stopping the process of providing the vacuum pressure to the vacuum adiabatic body, an operation of a vacuum pump connected to the vacuum space may be stopped. When inputting the getter, the process of heating or drying the vacuum adiabatic body may be performed together. Through this, the outgassing may be promoted. As another example, after the process of providing the getter to the vacuum adiabatic body is performed, the process of providing the vacuum pressure to the vacuum adiabatic body may be performed.

[0062] The time during which the vacuum adiabatic body vacuum exhaust process is performed may be referred to as a vacuum exhaust time. The vacuum exhaust time includes at least one of a time Δt_1 during which the process of heating or drying the vacuum adiabatic body is performed, a time Δt_2 during which the process of maintaining the getter in the vacuum adiabatic body is performed, of a time Δt_3 during which the process of cooling the vacuum adiabatic body is performed. Examples of times Δt_1 , Δt_2 , and Δt_3 are as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. In the vacuum adiabatic body vacuum exhaust process, the time Δt_1 may be a time t_{1a} or more and a time t_{1b} or less. As a first example, the time t_{1a} may be greater than or equal to about 0.2 hr and less than or equal to about 0.5 hr. The time t_{1b} may be greater than or equal to about 1 hr and less than or equal to about 24.0 hr. The time Δt_1 may be about 0.3 hr or more and about 12.0 hr or less. The time Δt_1 may be about 0.4 hr or more and about 8.0 hr or less. The time Δt_1 may be about 0.5 hr or more and about 4.0 hr or less. In this case, even if the Δt_1 is kept as short as possible, the sufficient outgassing may be applied to the vacuum adiabatic body. For example, this case may include a case in which a component of the vacuum adiabatic body, which is exposed to the vacuum space, among the components of the vacuum adiabatic body, has an outgassing rate (%) less than that of any one of the component of the vacuum adiabatic body, which is exposed to the external space of the vacuum space. Specifically, the component exposed to the vacuum space may include a portion having a outgassing rate less than that of a thermoplastic polymer. More specifically, the support or the radiation resistance sheet may be disposed in the vacuum space, and the outgassing rate of the support may be less than that of the thermoplastic plastic. As another example, this case may include a case in which a component of the vacuum adiabatic body, which is exposed to the vacuum space, among the components of the vacuum adiabatic body, has a max operating temperature ($^{\circ}\text{C.}$) greater than that of any one of the component of the vacuum adiabatic body, which is exposed to the external space of the vacuum space. In this case, the vacuum adiabatic body may be heated to a higher temperature to increase in outgassing rate. For example, the component exposed to the vacuum space may include a portion having an operating temperature greater than that of the thermoplastic polymer. As a more specific example, the support or the

radiation resistance sheet may be disposed in the vacuum space, and a use temperature of the support may be higher than that of the thermoplastic plastic. As another example, among the components of the vacuum adiabatic body, the component exposed to the vacuum space may contain more metallic portion than a non-metallic portion. That is, a mass of the metallic portion may be greater than a mass of the non-metallic portion, a volume of the metallic portion may be greater than a volume of the non-metallic portion, or an area of the metallic portion exposed to the vacuum space may be greater than an area exposed to the non-metallic portion of the vacuum space. When the components exposed to the vacuum space are provided in plurality, the sum of the volume of the metal material included in the first component and the volume of the metal material included in the second component may be greater than that of the volume of the non-metal material included in the first component and the volume of the non-metal material included in the second component. When the components exposed to the vacuum space are provided in plurality, the sum of the mass of the metal material included in the first component and the mass of the metal material included in the second component may be greater than that of the mass of the non-metal material included in the first component and the mass of the non-metal material included in the second component. When the components exposed to the vacuum space are provided in plurality, the sum of the area of the metal material, which is exposed to the vacuum space and included in the first component, and an area of the metal material, which is exposed to the vacuum space and included in the second component, may be greater than that of the area of the non-metal material, which is exposed to the vacuum space and included in the first component, and an area of the non-metal material, which is exposed to the vacuum space and included in the second component. As a second example, the time $t1a$ may be greater than or equal to about 0.5 hr and less than or equal to about 1 hr. The time $t1b$ may be greater than or equal to about 24.0 hr and less than or equal to about 65 hr. The time $\Delta t1$ may be about 1.0 hr or more and about 48.0 hr or less. The time $\Delta t1$ may be about 2 hr or more and about 24.0 hr or less. The time $\Delta t1$ may be about 3 hr or more and about 12.0 hr or less. In this case, it may be the vacuum adiabatic body that needs to maintain the $\Delta t1$ as long as possible. In this case, a case opposite to the examples described in the first example or a case in which the component exposed to the vacuum space is made of a thermoplastic material may be an example. A duplicated description will be omitted. In the vacuum adiabatic body vacuum exhaust process, the time $\Delta t1$ may be a time $t1a$ or more and a time $t1b$ or less. The time $t2a$ may be greater than or equal to about 0.1 hr and less than or equal to about 0.3 hr. The time $t2b$ may be greater than or equal to about 1 hr and less than or equal to about 5.0 hr. The time $\Delta t2$ may be about 0.2 hr or more and about 3.0 hr or less. The time $\Delta t2$ may be about 0.3 hr or more and about 2.0 hr or less. The time $\Delta t2$ may be about 0.5 hr or more and about 1.5 hr or less. In this case, even if the time $\Delta t2$ is kept as short as possible, the sufficient outgassing through the getter may be applied to the vacuum adiabatic body. In the vacuum adiabatic body vacuum exhaust process, the time $\Delta t3$ may be a time $t3a$ or more and a time $t3b$ or less. The time $t2a$ may be greater than or equal to about 0.2 hr and less than or equal to about 0.8 hr. The time $t2b$ may be greater than or equal to about 1 hr and less than or equal to about 65.0 hr. The time $\Delta t3$ may be about 0.2 hr or more and about 48.0 hr or less. The time $\Delta t3$ may be about 0.3 hr or more and about 24.0 hr or less. The time $\Delta t3$ may be about 0.4 hr or more and about 12.0 hr or less. The time $\Delta t3$ may be about 0.5 hr or more and about 5.0 hr or less. After the heating or drying process is performed during the exhaust process, the cooling process may be performed. For example, when the heating or drying process is performed for a long time, the time $\Delta t3$ may be long. The vacuum adiabatic body according to the present disclosure may be manufactured so that the time $\Delta t1$ is greater than the time $\Delta t2$, the time $\Delta t1$ is less than or equal to the time $\Delta t3$, or the time $\Delta t3$ is greater than the time $\Delta t2$. The following relational expression is satisfied: $\Delta t2 < \Delta t1 \leq \Delta t3$. The vacuum adiabatic body according to an embodiment may be manufactured so that the relational expression: $\Delta t1 + \Delta t2 + \Delta t3$ may be greater than or equal to about 0.3 hr and less than or equal to about 70 hr, be greater than or equal to about 1 hr and less than or equal to about 65 hr, or be greater than or equal to about 2 hr and less than or

equal to about 24 hr. The relational expression: $\Delta t_1 + \Delta t_2 + \Delta t_3$ may be manufactured to be greater than or equal to about 3 hr and less than or equal to about 6 hr.

[0063] An example of the vacuum pressure condition during the exhaust process is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. A minimum value of the vacuum pressure in the vacuum space during the exhaust process may be greater than about $1.8\text{E-}6$ Torr. The minimum value of the vacuum pressure may be greater than about $1.8\text{E-}6$ Torr and less than or equal to about $1.0\text{E-}4$ Torr, be greater than about $0.5\text{E-}6$ Torr and less than or equal to about $1.0\text{E-}4$ Torr, or be greater than about $0.5\text{E-}6$ Torr and less than or equal to about $0.5\text{E-}5$ Torr. The minimum value of the vacuum pressure may be greater than about $0.5\text{E-}6$ Torr and less than about $1.0\text{E-}5$ Torr. As such, the limitation in which the minimum value of the vacuum pressure provided during the exhaust process is because, even if the pressure is reduced through the vacuum pump during the exhaust process, the decrease in vacuum pressure is slowed below a certain level. As an embodiment, after the exhaust process is performed, the vacuum pressure of the vacuum space may be maintained at a pressure greater than or equal to about $1.0\text{E-}5$ Torr and less than or equal to about $5.0\text{E-}1$ Torr. The maintained vacuum pressure may be greater than or equal to about $1.0\text{E-}5$ Torr and less than or equal to about $1.0\text{E-}1$ Torr, be greater than or equal to about $1.0\text{E-}5$ Torr and less than or equal to about $1.0\text{E-}2$ Torr, be greater than or equal to about $1.0\text{E-}4$ Torr and less than or equal to about $1.0\text{E-}2$ Torr, or be greater than or equal to about $1.0\text{E-}5$ Torr and less than or equal to about $1.0\text{E-}3$ Torr. As a result of predicting the change in vacuum pressure with an accelerated experiment of two example products, one product may be provided so that the vacuum pressure is maintained below about $1.0\text{E-}04$ Torr even after about 16.3 years, and the other product may be provided so that the vacuum pressure is maintained below about $1.0\text{E-}04$ Torr even after about 17.8 years. As described above, the vacuum pressure of the vacuum adiabatic body may be used industrially only when it is maintained below a predetermined level even if there is a change over time.

[0064] FIG. 5a is a graph of an elapsing time and pressure in the exhaust process according to an example, and FIG. 5b is a view explaining results of a vacuum maintenance test in the acceleration experiment of the vacuum adiabatic body of the refrigerator having an internal volume of about 128 liters. Referring to FIG. 5b, it is seen that the vacuum pressure gradually increases according to the aging. For example, it is confirmed that the vacuum pressure is about $6.7\text{E-}04$ Torr after about 4.7 years, about $1.7\text{E-}03$ Torr after about 10 years, and about $1.0\text{E-}02$ Torr after about 59 years. According to these experimental results, it is confirmed that the vacuum adiabatic body according to the embodiment is sufficiently industrially applicable.

[0065] FIG. 6 is a graph illustrating results obtained by comparing the vacuum pressure with gas conductivity. Referring to FIG. 6, gas conductivity with respect to the vacuum pressure depending on a size of the gap in the vacuum space 50 was represented as a graph of effective heat transfer coefficient (eK). The effective heat transfer coefficient (eK) was measured when the gap in the vacuum space 50 has three values of about 3 mm, about 4.5 mm, and about 9 mm. The gap in the vacuum space 50 is defined as follows. When the radiation resistance sheet 32 exists inside surface vacuum space 50, the gap is a distance between the radiation resistance sheet 32 and the plate adjacent thereto. When the radiation resistance sheet 32 does not exist inside surface vacuum space 50, the gap is a distance between the first and second plates. It was seen that, since the size of the gap is small at a point corresponding to a typical effective heat transfer coefficient of about 0.0196 W/mK, which is provided to an adiabatic material formed by foaming polyurethane, the vacuum pressure is about $5.0\text{E-}1$ Torr even when the size of the gap is about 3 mm. Meanwhile, it was seen that the point at which reduction in adiabatic effect caused by the gas conduction heat is saturated even though the vacuum pressure decreases is a point at which the vacuum pressure is approximately $4.5\text{E-}3$ Torr. The vacuum pressure of about $4.5\text{E-}3$ Torr may be defined as the point at which the reduction in adiabatic effect caused by the gas conduction heat is saturated. Also, when the effective heat transfer coefficient is about 0.01 W/mK, the vacuum pressure is about

1.2E-2 Torr. An example of a range of the vacuum pressure in the vacuum space according to the gap is presented. The support may include at least one of a bar, a connection plate, or a support plate. In this case, when the gap of the vacuum space is greater than or equal to about 3 mm, the vacuum pressure may be greater than or equal to A and less than about 5E-1 Torr, or be greater than about 2.65E-1 Torr and less than about 5E-1 Torr. As another example, the support may include at least one of a bar, a connection plate, or a support plate. In this case, when the gap of the vacuum space is greater than or equal to about 4.5 mm, the vacuum pressure may be greater than or equal to A and less than about 3E-1 Torr, or be greater than about 1.2E-2 Torr and less than about 5E-1 Torr. As another example, the support may include at least one of a bar, a connection plate, or a support plate, and when the gap of the vacuum space is greater than or equal to about 9 mm, the vacuum pressure may be greater than or equal to A and less than about 1.0×10^{-1} Torr or be greater than about 4.5E-3 Torr and less than about 5E-1 Torr. Here, the A may be greater than or equal to about 1.0×10^{-6} Torr and less than or equal to about 1.0E-5 Torr. The A may be greater than or equal to about 1.0×10^{-5} Torr and less than or equal to about 1.0E-4 Torr. When the support includes a porous material or a filler, the vacuum pressure may be greater than or equal to about 4.7E-2 Torr and less than or equal to about 5E-1 Torr. In this case, it is understood that the size of the gap ranges from several micrometers to several hundreds of micrometers. When the support and the porous material are provided together in the vacuum space, a vacuum pressure may be created and used, which is middle between the vacuum pressure when only the support is used and the vacuum pressure when only the porous material is used.

[0066] FIG. 7 is a view illustrating various examples of the vacuum space. The present disclosure may be any one of the following examples or a combination of two or more examples.

[0067] Referring to FIG. 7, the vacuum adiabatic body according to the present disclosure may include a vacuum space. The vacuum space **50** may include a first vacuum space extending in a first direction (e.g., X-axis) and having a predetermined height. The vacuum space **50** may optionally include a second vacuum space (hereinafter, referred to as a vacuum space expansion portion) different from the first vacuum space in at least one of the height or the direction. The vacuum space expansion portion may be provided by allowing at least one of the first and second plates or the side plate to extend. In this case, the heat transfer resistance may increase by lengthening a heat conduction path along the plate. The vacuum space expansion portion in which the second plate extends may reinforce adiabatic performance of a front portion of the vacuum adiabatic body. The vacuum space expansion portion in which the second plate extends may reinforce adiabatic performance of a rear portion of the vacuum adiabatic body, and the vacuum space expansion portion in which the side plate extends may reinforce adiabatic performance of a side portion of the vacuum adiabatic body. Referring to FIG. 7a, the second plate may extend to provide the vacuum space expansion portion **51**. The second plate may include a second portion **202** extending from a first portion **201** defining the vacuum space **50** and the vacuum space expansion portion **51**. The second portion **202** of the second plate may branch a heat conduction path along the second plate to increase in heat transfer resistance. Referring to FIG. 7b, the side plate may extend to provide the vacuum space expansion portion. The side plate may include a second portion **152** extending from a first portion **151** defining the vacuum space **50** and the vacuum space extension portion **51**. The second portion of the side plate may branch the heat conduction path along the side plate to improve the adiabatic performance. The first and second portions **151** and **152** of the side plate may branch the heat conduction path to increase in heat transfer resistance. Referring to FIG. 7c, the first plate may extend to provide the vacuum space expansion portion. The first plate may include a second portion **102** extending from the first portion **101** defining the vacuum space **50** and the vacuum space expansion portion **51**. The second portion of the first plate may branch the heat conduction path along the second plate to increase in heat transfer resistance. Referring to FIG. 7d, the vacuum space expansion portion **51** may include an X-

direction expansion portion **51a** and a Y-direction expansion portion **51b** of the vacuum space. The vacuum space expansion portion **51** may extend in a plurality of directions of the vacuum space **50**. Thus, the adiabatic performance may be reinforced in multiple directions and may increase by lengthening the heat conduction path in the plurality of directions to improve the heat transfer resistance. The vacuum space expansion portion extending in the plurality of directions may further improve the adiabatic performance by branching the heat conduction path. Referring to FIG. **7e**, the side plate may provide the vacuum space extension portion extending in the plurality of directions. The vacuum space expansion portion may reinforce the adiabatic performance of the side portion of the vacuum adiabatic body. Referring to FIG. **7f**, the first plate may provide the vacuum space extension portion extending in the plurality of directions. The vacuum space expansion portion may reinforce the adiabatic performance of the side portion of the vacuum adiabatic body.

[0068] FIG. **8** is a view for explaining another adiabatic body. The present disclosure may be any one of the following examples or a combination of two or more examples. Referring to FIG. **8**, the vacuum adiabatic body according to the present disclosure may optionally include another adiabatic body **90**. Another adiabatic body may have a degree of vacuum less than that of the vacuum adiabatic body and be an object that does not include a portion having a vacuum state therein. The vacuum adiabatic body and another vacuum adiabatic body may be directly connected to each other or connected to each other through an intermedium. In this case, the intermedium may have a degree of vacuum less than that of at least one of the vacuum adiabatic body or another adiabatic body or may be an object that does not include a portion having the vacuum state therein. When the vacuum adiabatic body includes a portion in which the height of the vacuum adiabatic body is high and a portion in which the height of the vacuum adiabatic body is low, another adiabatic body may be disposed at a portion having the low height of the vacuum adiabatic body. Another adiabatic body may include a portion connected to at least a portion of the first and second plates and the side plate. Another adiabatic body may be supported on the plate or coupled or sealed. A degree of sealing between another adiabatic body and the plate may be lower than a degree of sealing between the plates. Another adiabatic body may include a cured adiabatic body (e.g., PU foaming solution) that is cured after being injected, a pre-molded resin, a peripheral adiabatic body, and a side panel. At least a portion of the plate may be provided to be disposed inside another adiabatic body. Another adiabatic body may include an empty space. The plate may be provided to be accommodated in the empty space. At least a portion of the plate may be provided to cover at least a portion of another adiabatic body. Another adiabatic body may include a member covering an outer surface thereof. The member may be at least a portion of the plate. Another adiabatic body may be an intermedium for connecting, supporting, bonding, or sealing the vacuum adiabatic body to the component. Another adiabatic body may be an intermedium for connecting, supporting, bonding, or sealing the vacuum adiabatic body to another vacuum adiabatic body. Another adiabatic body may include a portion connected to a component coupling portion provided on at least a portion of the plate. Another adiabatic body may include a portion connected to a cover covering another adiabatic body. The cover may be disposed between the first plate and the first space, between the second plate and the second space, or between the side plate and a space other than the vacuum space **50**. For example, the cover may include a portion on which the component is mounted. As another example, the cover may include a portion that defines an outer appearance of another adiabatic body. Referring to FIGS. **8a** to **8f**, another adiabatic body may include a peripheral adiabatic body. The peripheral adiabatic body may be disposed on at least a portion of a periphery of the vacuum adiabatic body, a periphery of the first plate, a periphery of the second plate, and the side plate. The peripheral adiabatic body disposed on the periphery of the first plate or the periphery of the second plate may extend to a portion at which the side plate is disposed or may extend to the outside of the side plate. The peripheral adiabatic body disposed on the side plate may extend to a portion at which the first plate or may extend to the outside of the first plate or the second plate. Referring to FIGS. **8g** to **8h**, another adiabatic body may include a

central adiabatic body. The central adiabatic body may be disposed on at least a portion of a central portion of the vacuum adiabatic body, a central portion of the first plate, or a central portion of the second plate.

[0069] Referring to FIG. **8a**, the peripheral adiabatic body **92** may be placed on the periphery of the first plate. The peripheral adiabatic body may be in contact with the first plate. The peripheral adiabatic body may be separated from the first plate or further extend from the first plate (indicated by dotted lines). The peripheral adiabatic body may improve the adiabatic performance of the periphery of the first plate. Referring to FIG. **8b**, the peripheral adiabatic body may be placed on the periphery of the second plate. The peripheral adiabatic body may be in contact with the second plate. The peripheral adiabatic body may be separated from the second plate or further extend from the second plate (indicated by dotted lines). The peripheral adiabatic body may improve the adiabatic performance of the periphery of the second plate. Referring to FIG. **8c**, the peripheral adiabatic body may be disposed on the periphery of the side plate. The peripheral adiabatic body may be in contact with the side plate. The peripheral adiabatic body may be separated from the side plate or further extend from the side plate. The peripheral adiabatic body may improve the adiabatic performance of the periphery of the side plate. Referring to FIG. **8d**, the peripheral adiabatic body **92** may be disposed on the periphery of the first plate. The peripheral adiabatic body may be placed on the periphery of the first plate constituting the vacuum space expansion portion **51**. The peripheral adiabatic body may be in contact with the first plate constituting the vacuum space extension portion. The peripheral adiabatic body may be separated from or further extend to the first plate constituting the vacuum space extension portion. The peripheral adiabatic body may improve the adiabatic performance of the periphery of the first plate constituting the vacuum space expansion portion. Referring to FIGS. **8e** and **8f**, in the peripheral adiabatic body, the vacuum space extension portion may be disposed on a periphery of the second plate or the side plate. The same explanation as in FIG. **8d** may be applied. Referring to FIG. **8g**, the central adiabatic body **91** may be placed on a central portion of the first plate. The central adiabatic body may improve adiabatic performance of the central portion of the first plate. Referring to FIG. **8h**, the central adiabatic body may be disposed on the central portion of the second plate. The central adiabatic body may improve adiabatic performance of the central portion of the second plate.

[0070] FIG. **9** is a view for explaining a heat transfer path between first and second plates having different temperatures. An example of the heat transfer path is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples.

[0071] The heat transfer path may pass through the extension portion at at least a portion of the first portion **101** of the first plate, the first portion **201** of the second plate, or the first portion **151** of the side plate. The first portion may include a portion defining the vacuum space. The extension portions **102**, **152**, and **202** may include portions extending in a direction away from the first portion. The extension portion may include a side portion of the vacuum adiabatic body, a side portion of the plate having a higher temperature among the first and second plates, or a portion extending toward the side portion of the vacuum space **50**. The extension portion may include a front portion of the vacuum adiabatic body, a front portion of the plate having a higher temperature among the first and second plates, or a front portion extending in a direction away from the front portion of the vacuum space **50**. Through this, it is possible to reduce generation of dew on the front portion. The vacuum adiabatic body or the vacuum space **50** may include first and second surfaces having different temperatures from each other. The temperature of the first surface may be lower than that of the second surface. For example, the first surface may be the first plate, and the second surface may be the second plate. The extension portion may extend in a direction away from the second surface or include a portion extending toward the first surface. The extension portion may include a portion, which is in contact with the second surface, or a portion extending in a state of being in contact with the second surface. The extension portion may include a portion extending to be spaced apart from the two surfaces. The extension portion may include a portion

having heat transfer resistance greater than that of at least a portion of the plate or the first surface. The extension portion may include a plurality of portions extending in different directions. For example, the extension portion may include a second portion **202** of the second plate and a third portion **203** of the second plate. The third portion may also be provided on the first plate or the side plate. Through this, it is possible to increase in heat transfer resistance by lengthening the heat transfer path. In the extension portion, the above-described heat transfer resistor may be disposed. Another adiabatic body may be disposed outside the extending portion. Through this, the extension portion may reduce generation of dew on the second surface. Referring to FIG. **9a**, the second plate may include the extension portion extending to the periphery of the second plate. Here, the extension portion may further include a portion extending backward. Referring to FIG. **9b**, the side plate may include the extension portion extending to a periphery of the side plate. Here, the extension portion may be provided to have a length that is less than or equal to that of the extension portion of the second plate. Here, the extension portion may further include a portion extending backward. Referring to FIG. **9c**, the first plate may include the extension portion extending to the periphery of the first plate. Here, the extension portion may extend to a length that is less than or equal to that of the extension portion of the second plate. Here, the extension portion may further include a portion extending backward.

[0072] FIG. **10** is a view for explaining a branch portion on the heat transfer path between first and second plates having different temperatures. An example of the branch portion is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples.

[0073] Optionally, the heat transfer path may pass through portions **205**, **153**, and **104**, each of which is branched from at least a portion of the first plate, the second plate, or the side plate. Here, the branched heat transfer path means a heat transfer path through which heat flows to be separated in a different direction from the heat transfer path through which heat flows along the plate. The branched portion may be disposed in a direction away from the vacuum space **50**. The branched portion may be disposed in a direction toward the inside of the vacuum space **50**. The branched portion may perform the same function as the extension portion described with reference to FIG. **9**, and thus, a description of the same portion will be omitted. Referring to FIG. **10a**, the second plate may include the branched portion **205**. The branched portion may be provided in plurality, which are spaced apart from each other. The branched portion may include a third portion **203** of the second plate. Referring to FIG. **10b**, the side plate may include the branched portion **153**. The branched portion **153** may be branched from the second portion **152** of the side plate. The branched portion **153** may provide at least two. At least two branched portions **153** spaced apart from each other may be provided on the second portion **152** of the side plate. Referring to FIG. **10c**, the first plate may include the branched portion **104**. The branched portion may further extend from the second portion **102** of the first plate. The branched portion may extend toward the periphery. The branched portion **104** may be bent to further extend. A direction in which the branched portion extends in FIGS. **10a**, **10b**, and **10c** may be the same as at least one of the extension directions of the extension portion described in FIG. **10**.

[0074] FIG. **11** is a view for explaining a process of manufacturing the vacuum adiabatic body.

[0075] Optionally, the vacuum adiabatic body may be manufactured by a vacuum adiabatic body component preparation process in which the first plate and the second plate are prepared in advance. Optionally, the vacuum adiabatic body may be manufactured by a vacuum adiabatic body component assembly process in which the first plate and the second plate are assembled.

Optionally, the vacuum adiabatic body may be manufactured by a vacuum adiabatic body vacuum exhaust process in which a gas in the space defined between the first plate and the second plate is discharged. Optionally, after the vacuum adiabatic body component preparation process is performed, the vacuum adiabatic body component assembly process or the vacuum adiabatic body exhaust process may be performed. Optionally, after the vacuum adiabatic body component

assembly process is performed, the vacuum adiabatic body vacuum exhaust process may be performed. Optionally, the vacuum adiabatic body may be manufactured by the vacuum adiabatic body component sealing process (S3) in which the space between the first plate and the second plate is sealed. The vacuum adiabatic body component sealing process may be performed before the vacuum adiabatic body vacuum exhaust process (S4). The vacuum adiabatic body may be manufactured as an object with a specific purpose by an apparatus assembly process (S5) in which the vacuum adiabatic body is combined with the components constituting the apparatus. The apparatus assembly process may be performed after the vacuum adiabatic body vacuum exhaust process. Here, the components constituting the apparatus means components constituting the apparatus together with the vacuum adiabatic body.

[0076] The vacuum adiabatic body component preparation process (S1) is a process in which components constituting the vacuum adiabatic body are prepared or manufactured. Examples of the components constituting the vacuum adiabatic body may include various components such as a plate, a support, a heat transfer resistor, and a tube. The vacuum adiabatic body component assembly process (S2) is a process in which the prepared components are assembled. The vacuum adiabatic body component assembly process may include a process of disposing at least a portion of the support and the heat transfer resistor on at least a portion of the plate. For example, the vacuum adiabatic body component assembly process may include a process of disposing at least a portion of the support and the heat transfer resistor between the first plate and the second plate. Optionally, the vacuum adiabatic body component assembly process may include a process of disposing a penetration component on at least a portion of the plate. For example, the vacuum adiabatic body component assembly process may include a process of disposing the penetration component or a surface component between the first and second plates. After the penetration component may be disposed between the first plate and the second plate, the penetration component may be connected or sealed to the penetration component coupling portion.

[0077] An example of a vacuum adiabatic body vacuum exhaust process vacuum is as follows. The present disclosure may be any one of the, examples or a combination of two or more examples. The vacuum adiabatic body vacuum exhaust process may include at least one of a process of inputting the vacuum adiabatic body into an exhaust passage, a getter activation process, a process of checking vacuum leakage and a process of closing the exhaust port. The process of forming the coupling part may be performed in at least one of the vacuum adiabatic body component preparation process, the vacuum adiabatic body component assembly process, or the apparatus assembly process. Before the vacuum adiabatic body exhaust process is performed, a process of washing the components constituting the vacuum adiabatic body may be performed. Optionally, the washing process may include a process of applying ultrasonic waves to the components constituting the vacuum adiabatic body or a process of providing ethanol or a material containing ethanol to surfaces of the components constituting the vacuum adiabatic body. The ultrasonic wave may have an intensity between about 10 kHz and about 50 kHz. A content of ethanol in the material may be about 50% or more. For example, the content of ethanol in the material may range of about 50% to about 90%. As another example, the content of ethanol in the material may range of about 60% to about 80%. As another example, the content of ethanol in the material may be range of about 65% to about 75%. Optionally, after the washing process is performed, a process of drying the components constituting the vacuum adiabatic body may be performed. Optionally, after the washing process is performed, a process of heating the components constituting the vacuum adiabatic body may be performed.

[0078] The contents described in FIGS. 1 to 11 may be applied to all or selectively applied to the embodiments described with reference to the drawings below.

[0079] As an embodiment, an example of a process associated with a heat transfer resistor is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. The vacuum adiabatic body component preparation process may include a

process of manufacturing the heat transfer resistor. Before the vacuum adiabatic body vacuum exhaust process is performed, the process of manufacturing the heat transfer resistor may be performed. The heat transfer resistor may be manufactured by a metal sheet. Optionally, before the vacuum adiabatic body vacuum exhaust process is performed, the process of washing the heat transfer resistor may be performed. Optionally, before the vacuum adiabatic body vacuum exhaust process is performed, a process of providing the component coupling portion to a portion of the heat transfer resistor may be performed. Optionally, the process of providing the component coupling portion may include a process of manufacturing a tube provided to the component coupling portion. The tube may be connected to a portion of the heat transfer resistor. The tube may be disposed in an empty space provided in the heat transfer resistor or in an empty space provided between the heat transfer resistors. Optionally, the process of providing the component coupling portion may include a process of providing a through-hole in a portion of the heat transfer resistor. Optionally, the process of providing the component coupling portion may include a process of providing a curved portion to at least one of the heat transfer resistor or the tube.

[0080] Optionally, during the vacuum adiabatic body vacuum exhaust process is performed, the process of deforming the heat transfer resistor may be performed. An example of the process of deforming the heat transfer resistor may be applied to the process of deforming the plate. An example of the process of deforming the heat transfer resistor may be applied when at least a portion of the plate and the heat transfer resistor are integrated with each other. Examples of the process of deforming the heat transfer resistor are as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. First, the process may include a process in which the heat transfer resistor is recessed in a direction toward the internal space or toward the outside of the internal space. Therefore, a heat transfer path may extend to reduce a degree of heat conduction through the heat transfer resistor. Second, the process may include a process of changing a curvature radius of the heat transfer resistor. For example, the process of changing the curvature radius may include changing the curvature radius in at least a portion of a central portion and a peripheral portion of the heat transfer resistor. As another example, the process of changing the curvature radius may include a process of changing the curvature radius in the vicinity of an empty space provided inside the support, or a process of changing the curvature radius in the vicinity of an empty space provided to the outside of an edge of the support. As another example, the process of deforming the curvature radius may include a process of providing the heat transfer resistor to a portion, which is not in contact with the support. As another example, the process of changing the curvature radius may include a process of changing a curvature radius at at least a portion of the first portion or the second portion of the heat transfer resistor. Here, the first portion of the heat transfer resistor may be a portion defining a vacuum space. The second portion of the heat transfer resistor may be a portion extending in a direction away from the first portion to the vacuum space. Third, the process may include a process of changing a thickness of the heat transfer resistor. For example, the process of changing the thickness may include a process of changing a thickness at the portion supported by the support. As another example, the process of changing the thickness may include a process of changing a thickness in the vicinity of the empty space provided inside the support. As another example, the process of changing the thickness may include a process of changing a thickness in the vicinity of the empty space provided outside the edge of the support. As another example, the process of changing the thickness may include a process of providing the heat transfer resistor to a portion that is not in contact with the support. After the process of changing the curvature radius or the thickness in the central portion of the heat transfer resistor is performed, the process of changing the curvature radius or the thickness in the peripheral portion of the heat transfer resistor may be performed. After the process of changing the curvature radius or the thickness is performed in the vicinity of the empty space provided inside the support, the process of changing the curvature radius in the vicinity of the empty space provided outside the edge of the support may be

performed. After the process of changing the curvature radius or the thickness in the first portion of the heat transfer resistor is performed, the process of changing the curvature radius or the thickness in the second portion of the heat transfer resistor may be performed. The process of deforming the heat transfer resistor while the vacuum adiabatic body exhaust process is performed may also be applied to the plate, and the same description will be omitted.

[0081] The process associated with the heat transfer resistor may optionally include a process related to the process of washing the heat transfer resistor. An example of a process sequence associated with the process of washing the heat transfer resistor is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. After the process of manufacturing the heat transfer resistor is performed, at least one of the process of manufacturing the heat transfer resistor and the process of washing the heat transfer resistor may be performed. After the process of manufacturing the heat transfer resistor is performed, the process of washing the heat transfer resistor may be performed. Before the process of manufacturing the heat transfer resistor is performed, the process of washing the heat transfer resistor may be performed. After the process of manufacturing the heat transfer resistor is performed, at least one of a process of providing the component coupling portion to a portion of the heat transfer resistor or the process of washing the heat transfer resistor may be performed. After the process of providing the component coupling portion to a portion of the heat transfer resistor is performed, the process of washing the heat transfer resistor may be performed.

[0082] The process associated with the heat transfer resistor may optionally include a process related to the process of providing the component coupling portion to the heat transfer resistor. An example of a process sequence related to the process of providing the component coupling portion to the heat transfer resistor is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. After the process of providing the through-hole in a portion of the heat transfer resistor is performed, at least one of a process of providing a curved portion to the heat transfer resistor, a process of providing a curved portion to the tube, or a process of providing a seal between the heat transfer resistor and the tube may be performed. After the process of providing the curved portion to at least a portion of at least one of the heat transfer resistor or the tube is performed, the process of sealing the gap between the plate and the tube may be performed. The process of providing the through-hole in the portion of the heat transfer resistor and the process of providing the curved portion on at least one of the heat transfer resistor or the tube may be performed at the same time. The process of providing the through-hole in a portion of the heat transfer resistor and the process of providing the seal between the heat transfer resistor and the tube may be performed at the same time. After the process of providing the curved portion to the tube connected to the heat transfer resistor is performed, the process of providing the through-hole in the portion of the heat transfer resistor may be performed. Before the vacuum adiabatic body vacuum exhaust process is performed, a portion of the tube may be provided and/or sealed to the heat transfer resistor, and after the vacuum adiabatic body vacuum exhaust process is performed, the other portion of the tube may be sealed.

[0083] The process associated with the heat transfer resistor may optionally include a process related to the process of providing the heat transfer resistor on at least a portion of the plate. An example of a process sequence related to the process of providing the heat transfer resistor to at least a portion of the plate is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. Before the vacuum adiabatic body exhaust process is performed, the heat transfer resistor may be connected to at least one of the first plate or the second plate. Before the vacuum adiabatic body exhaust process is performed, the heat transfer resistor may be disposed in a heat conduction path, through which a fluid flows along the internal space. Before the vacuum adiabatic body exhaust process is performed, the heat transfer resistor may be provided in a space between the first plate and the second plate. Before the vacuum adiabatic body exhaust process is performed, the heat transfer resistor may be provided at the inside

of the plate or the surface of the plate. Before the vacuum adiabatic body exhaust process is performed, the heat transfer resistor may be disposed to be supported by at least a portion of the plate. Before the vacuum adiabatic body vacuum exhaust process is performed, the heat transfer resistor may be disposed to be supported by the support.

[0084] When at least a portion of the heat transfer resistor is used to be integrated with the plate, the example of the process associated with the heat transfer resistor may also be applied to the example of the process of the plate.

[0085] Referring to FIG. 12, the vacuum adiabatic body according to the present invention includes a heat transfer path formed between the plates having different temperatures, and optionally, the heat transfer path may include a portion passing through the heat transfer resistor. An example of the heat transfer resistor as the heat transfer path is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. The heat transfer resistor may be provided to be integrated with at least one of the first or second plate. The heat transfer resistor may be provided to be integrated with any one of the first and second plates. The heat transfer resistor may be provided as any one of the first and second plates. The heat transfer resistor may be provided as a portion of any one of the first and second plates. The heat transfer resistor may be provided as a component separated from the other one of the first and second plates. In this case, optionally, the heat transfer resistor may be provided to be coupled or sealed to the other one of the first and second plates. The heat transfer resistor may include a portion having a degree of heat transfer resistance, which is greater than that of at least a portion of the other one of the first and second plates. The heat transfer resistor may include a portion having a thickness less than that of at least a portion of the other one of the first and second plates. The heat transfer resistor may include a portion having a curvature radius less than that of at least a portion of the other one of the first and second plates. The heat transfer resistor may include a portion having a curvature radius less than that of at least a portion of the other one of the first and second plates.

[0086] Referring to FIG. 12a, the heat transfer resistor 60 may be provided on the first plate 10. The heat transfer resistor may be at least one of a radiation resistance sheet, a porous material, a filler, or a conductive resistance sheet. More preferably, the heat transfer resistor may be the conduction resistance sheet. A shield portion for thermal insulation or a member for reinforcing strength may be provided on an outer surface of the heat transfer resistor. The heat transfer resistor may be installed in two opposite peripheral portions of the vacuum space 50. The heat transfer resistor may be installed to be connected to two opposite edges of the vacuum space. Referring to FIG. 12b, the heat transfer resistor may be provided on the side plate. Referring to FIG. 12c, the heat transfer resistor may be provided on the second plate. In FIGS. 12b and 12c, the relationship between the plate and the heat transfer resistor is the same as that of FIG. 12a. Referring to FIG. 12d, the heat transfer resistor may be provided to be integrated with the first plate. In this case, the heat transfer resistor may be provided as the first plate or may be provided as a portion of the first plate. Referring to FIG. 12e, the heat transfer resistor may be provided to be integrated with the side plate. In this case, the heat transfer resistor may be provided as the side plate or as a portion of the first plate. Referring to FIG. 12f, the heat transfer resistor may be provided to be integrated with the second plate. In this case, the heat transfer resistor may be provided as the second plate or as a portion of the second plate.

[0087] The vacuum adiabatic body according to the present invention includes a heat transfer path formed between the plates having different temperatures, and optionally, the heat transfer path may include a portion passing through the side plate. An example of the side plate as the heat transfer path is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. The side plate may be provided to be integrated with at least one of the first or second plate. The side plate may be provided to be integrated with any one of the first and second plates. The side plate may be provided as any one of the first and second plates. The side plate may be provided as a portion of any one of the first and second plates. The side plate

may be provided as a component separated from the other of the first and second plates. In this case, optionally, the side plate may be provided to be coupled or sealed to the other one of the first and second plates. The side plate may include a portion having a degree of strain resistance, which is greater than that of at least a portion of the other one of the first and second plates. The side plate may include a portion having a thickness greater than that of at least a portion of the other one of the first and second plates. The side plate may include a portion having a curvature radius less than that of at least a portion of the other one of the first and second plates. FIGS. 12c and 12f may illustrate an example in which the first plate and the side plate are integrally provided. FIGS. 12a and 12d may illustrate an example in which the second plate and the side plate are integrally provided. FIGS. 12b and 12e may illustrate an example in which the side plate is provided as a separate component that is separated from the first plate and the second plate.

[0088] As an embodiment, an example of a process associated with the support is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. The vacuum adiabatic body component preparation process may include a process of manufacturing the support. Before the vacuum adiabatic body vacuum exhaust process is performed, the process of manufacturing the support may be performed. For example, the support may be manufactured through the injection. Optionally, before the vacuum adiabatic body vacuum exhaust process is performed, the process of washing the support may be performed. Before the vacuum adiabatic body vacuum exhaust process is performed or while the vacuum adiabatic body vacuum exhaust process is performed, a process of storing the support under a predetermined condition may be performed. For example, before the vacuum adiabatic body vacuum exhaust process is performed, a primary storage process may be performed, and while the vacuum adiabatic body vacuum exhaust process is performed, a secondary storage process may be performed. For another example, during the vacuum adiabatic body vacuum exhaust process is performed, the storage process may be performed. Examples of the storage process are as follows. As a first example, the storage process may include a process of drying or heating the support. Thus, the outgassing from the support may be performed. The heating temperature may be greater than a predetermined reference temperature and less than a melting point of the support. The predetermined reference temperature may be a temperature between about 10 degrees and about 40 degrees. The heating temperature may be greater than about 80 degrees and less than about 280 degrees. The heating temperature may be greater than about 100 degrees and less than about 260 degrees. The heating temperature may be greater than about 120 degrees and less than about 240 degrees. The heating temperature may be greater than about 140 degrees and less than about 220 degrees. The heating temperature may be greater than about 160 degrees and less than about 200 degrees. The heating temperature may be greater than about 170 degrees and less than about 190 degrees. The heating temperature in the primary storage process may be less than the heating temperature in the secondary storage process. Optionally, the storage process may include a process of cooling the support. After the process of drying or heating the support is performed, the process of cooling the support may be performed. As a second example, the storage process may include a process of storing the support in a state of a temperature less than atmospheric pressure. Thus, the outgassing from the support may be performed. The storage pressure may be less than a pressure in a vacuum state in which the internal space between the first plate and the second plate is maintained. The storage pressure may be greater than 10E-10 torr and less than atmospheric pressure. The storage pressure may be greater than 10E-9 torr and less than atmospheric pressure. The storage pressure may be greater than 10E-8 torr and less than atmospheric pressure. The storage pressure may be greater than 10E-7 torr and less than atmospheric pressure. The storage pressure may be in a state of being greater than 10E-3 torr and less than atmospheric pressure. The storage pressure may be in a state of being greater than 10E-2 torr and less than atmospheric pressure. The storage pressure may be in a state of being greater than 0.5E-1 torr and less than atmospheric pressure. The storage pressure may be in a state of being greater than 0.5E-1 torr and

less than $3E-1$ torr. The storage pressure in the primary storage process may be higher than the storage pressure in the secondary storage process. Optionally, the storage process may include a storage process at the atmospheric pressure. After the process of storing the support in a state of the pressure less than the atmospheric pressure is performed, the process of storing the support in the state of the atmospheric pressure may be performed.

[0089] Optionally, before the vacuum adiabatic body vacuum exhaust process is performed, a process of coupling a plurality of portions of the support to each other may be performed. For example, the coupling process may include a process of coupling a bar of the support to a connection plate. As another example, the coupling process may include a process of coupling the bar of the support to the support plate.

[0090] The process associated with the support may optionally include a process related to the process of storing the support under the predetermined condition. An example of a process sequence related to the process in which the support is stored under the predetermined condition is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. After the process of drying or heating the support is performed, at least one of the process of storing the support at the temperature less than atmospheric pressure, the process of cooling the support, or the process of storing the support at the atmospheric pressure may be performed. After the process of storing the support at the pressure less than the atmospheric pressure is performed, at least one of the process of drying or heating the support, the process of cooling the support, or the process of storing the support at the atmospheric pressure may be performed. The process of drying or heating the support and the process of storing the support at the pressure less than the atmospheric pressure may be performed at the same time. The process of drying or heating the support and the process of storing the support at the atmospheric pressure may be performed at the same time. The process of storing the support under the condition less than atmospheric pressure and the process of cooling the support may be performed at the same time.

[0091] The process associated with the support may optionally include a process related to the process in which the support is coupled. An example of a process sequence related to the process in which the support is coupled is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. Before the coupling process is performed, a process of providing a separate component separated from the support in a space provided inside the support may be performed. For example, the component may include a heat transfer resistor. After the coupling process is performed, the support may be packaged or stored in a vacuum state. After the process of storing the support under the predetermined condition is performed, a process of coupling a plurality of portions of the support to each other may be performed.

[0092] In relation to the support, the process may optionally include a process related to the process of washing the support. An example of a process sequence related to the process of washing the support is as follows. The present disclosure may be any one of the following examples or a combination of two or more examples. After the process of manufacturing the support is performed, at least one of the process of washing the support, the process of storing the support under the predetermined condition, or the process of coupling the plurality of portions of the support to each other may be performed. After the process of washing the support is performed, at least one of the process of storing the support under the predetermined condition or the process of coupling the plurality of portions of the support to each other may be performed. Before the process of washing the support is performed, at least one of the process of storing the support under the predetermined condition or the process of coupling the plurality of portions of the support to each other may be performed.

[0093] The process associated with the support may optionally include a process related to the process of providing the support to plate. An example of a process sequence related to the process of providing the support to the plate is as follows. The present disclosure may be any one of the

following examples or a combination of two or more examples. Before the vacuum adiabatic body exhaust process is performed, the support may be provided in a space between the first plate and the second plate. Before the vacuum adiabatic body exhaust process is performed, the support may be provided at the inside of the plate or the surface of the plate. Before the vacuum adiabatic body vacuum exhaust process is performed, the support may be coupled to the plate. After the component coupling portion is provided on a portion of the plate, the support may be provided in the space between the first plate and the second plate.

[0094] FIG. 13 is a view for explaining deformation and displacement of a third straight portion.

[0095] Referring to FIG. 13, the side plate 15 may be deformed when a vacuum pressure is applied to the vacuum space 50. The side plate 15 may be bent using the second curved portion 232 as a supporting point by the vacuum pressure. The first plate 10 is thinner than the second plate 20 and has weaker strength. It is difficult for the first plate 10 to support the bending of the side plate 15. When the side plate 15 is bent using the second curved portion 232 as a supporting point, the third straight portion 223 and the support 30 may be in contact with each other. When the side plate 15 and the support 30 are in contact with each other to support each other, strength of the peripheral portion of the vacuum adiabatic body may increase. This is because the two members of the side plate 15 and the support 30 act as a single body.

[0096] There is a possibility of a limitation in that heat conduction increases due to the contact between the support and the side plate. For example, heat conducted through the first plate 10 and the support 30 may be conducted through the side plate 15. Since the support 30 may be made of a resin material, and the side plate 15 may be made of stainless steel, the increase in conductive heat may be greater. Hot air or cold air through the side plate 15 may be conducted to the second plate 20. When the first plate 10 is under a low temperature environment, and the second plate 20 is under a high temperature environment, a surface of the second plate 20 may have a low temperature. Dew condensation may occur on the low-temperature surface of the second plate 20.

[0097] A configuration for preventing the dew formation is proposed in the embodiment.

[0098] At least a portion of the third straight portion 223 may be in contact with the support 30. At least a portion of the third straight portion 223 may be in contact with the support 30 after the vacuum adiabatic body component assembly process is performed. A portion of the third straight portion 223 that is in contact with the support 30 may be an intermediate portion of the third straight portion 223. An upper end and a lower end of the third straight portion 223 may not be in contact with the support 30. According to the above configuration, the conductive heat between the third straight portion 223 and the support 30 may be reduced.

[0099] The first straight portion 221 may not be in contact with the support 30 after the vacuum adiabatic body component assembly process is performed. When the first straight portion 221 and the support 30 are not in contact with each other, the first plate 10, which is a thin plate, may be placed between the two members. The first plate 10 may serve as a conductive resistance sheet.

[0100] At least a portion of the plurality of curved portions provided in the vacuum adiabatic body may not be supported by the support 30 after the vacuum adiabatic body component assembling process is performed. At least a portion of the first curved portion 231 may not be in contact with the support 30. Thus, it is possible to prevent the severe cold air at a top end of the support 30 from being conducted to the side plate 15 made of a metal material. Cold air that is capable of being conducted riding over the side plate 15 may be reduced. At least a portion of the second curved portion 232 may not be in contact with the support 30. Thus, the conduction heat may be reduced by the support 30 and the side plate 15.

[0101] At least a portion of the plurality of curved portions provided in the vacuum adiabatic body may be supported by the support 30 after the vacuum adiabatic body component assembling process is performed. At least a portion of the second curved portion 232 may be in contact with the support 30. Thus, the support 30 may support the side plate 15. When the second curved portion 232 is in contact with the support 30, the side plate 15 and the support 30 are in close contact with

each other by the vacuum pressure to move as a single body. Accordingly, the strength may increase.

[0102] FIG. **13** is a view illustrating a state before the component assembly process of the vacuum adiabatic body is performed, and FIG. **16** is a view illustrating a state after the component assembly process of the vacuum adiabatic body is performed.

[0103] FIGS. **13** and **16** will be compared to be described.

[0104] At least a portion of the third straight portion **223** may be in contact with the support **30**. At least a portion of the third straight portion **223** may be in contact with the support **30** after the vacuum adiabatic body component assembly process is performed. A central portion of the third straight portion **223** may be in contact with the support **30**. A minimum area of the two members (the third straight portion and the support) may be in contact with each other for strength reinforcement. To prevent the heat conduction, upper portions of the two members (the third straight portion and the support) may not be in contact with each other. To prevent the heat conduction, lower portions of the two members (the third straight portion and the support) may not be in contact with each other. Accordingly, the heat conduction between the two members may be reduced. Dew formation of the second plate may be reduced.

[0105] The first straight portion **221** may not be in contact with the support **30** before the vacuum adiabatic body component assembling process is performed. The first straight portion **221** may not be in contact with the support **30** after the vacuum adiabatic body component assembly process is performed. When the first straight portion **221** and the support **30** are not in contact with each other, the first plate **10**, which is a thin plate, may be placed between the two members. The thin plate may serve as a conductive resistance sheet. The first plate **10** may be the thin plate. The thin plate may be thinner than each of other members. The thin plate is difficult to allow heat to pass therethrough. The cold air or hot air may not be transmitted through the thin plate.

[0106] At least a portion of the plurality of curved portions provided in the vacuum adiabatic body may not be supported by the support **30** before the vacuum adiabatic body component assembling process is performed. At least a portion of the plurality of curved portions provided in the vacuum adiabatic body may not be supported by the support **30** after the vacuum adiabatic body component assembling process is performed. At least a portion of the first curved portion **231** may not be in contact with the support **30** before the vacuum adiabatic body component assembling process is performed. At least a portion of the first curved portion **231** may not be in contact with the support **30** after the vacuum adiabatic body component assembly process is performed. Thus, it is possible to prevent the severe cold air at an top end of the support **30** from being conducted to the side plate **15** made of a metal material. Cold air that is capable of being conducted riding over the side plate **15** may be reduced.

[0107] At least a portion of the second curved portion **232** may not be in contact with the support **30** before the vacuum adiabatic body component assembling process is performed. At least a portion of the second curved portion **232** may not be in contact with the support **30** after the vacuum adiabatic body component assembly process is performed. Thus, the conduction heat may be reduced by the support **30** and the side plate **15**.

[0108] At least a portion of the plurality of curved portions provided in the vacuum adiabatic body may not be supported by the support **30** before the vacuum adiabatic body component assembling process is performed. At least a portion of the plurality of curved portions provided in the vacuum adiabatic body may be supported by the support **30** after the vacuum adiabatic body component assembling process is performed.

[0109] At least a portion of the second curved portion **232** may not be supported by the support **30** before the vacuum adiabatic body component assembly process is performed. At least a portion of the second curved portion **232** may be in contact with the support **30** after the vacuum adiabatic body component assembly process is performed. Thus, the support **30** may support the side plate **15**. When the second curved portion **232** is in contact with the support **30**, the side plate **15** and the

support **30** are in close contact with each other by the vacuum pressure to move as a single body. Accordingly, the strength may increase.

[0110] FIG. **14** is a view for explaining the peripheral portion of the radiation resistance sheet.

[0111] The radiation resistance sheet **32** is a member lying on a plane in the longitudinal direction of the space in the vacuum state. The radiation resistance sheet **32** may have a sheet shape, and the support **30** may support the radiation resistance sheet **32** at at least two positions spaced apart from each other. At least one radiation resistance sheet **32** may be provided. When at least two or more radiation resistance sheets **32** are placed, the radiation resistance sheets **32** may be placed in parallel to each other.

[0112] Referring to FIG. **14**, after the vacuum adiabatic body component assembly process is performed, the radiation resistance sheet **32** may not be in contact with even any one of the first plate **10**, the second plate **20**, and the side plate **15**. After the vacuum adiabatic body component assembly process is performed, the radiation resistance sheet **32** may not be in contact with the first plate **10**. After the vacuum adiabatic body component assembly process is performed, the radiation resistance sheet **32** may not be in contact with the second plate **20**. After the vacuum adiabatic body component assembly process is performed, the radiation resistance sheet **32** may not be in contact with the side plate **15**. The heat conduction may not occur by blocking the contact between the radiation resistance sheet **32** made of aluminum and the plate made of stainless steel. It is possible to increase in adiabatic efficiency of the vacuum adiabatic body.

[0113] An edge of the radiation resistance sheet **32** may be placed at a gap between a first bar **319** and a second bar **318**. The first bar may be the outermost bar. The second bar may be adjacent to a center of the vacuum space **50** from the first bar. The radiation resistance sheet **32** may extend to a gap between the first bar and the second bar. The edge of the radiation resistance sheet **32** may not be placed in a gap between the first straight portion **221** and the first bar. Accordingly, it is possible to block the contact between the radiation resistance sheet **32** and the plate.

[0114] The radiation resistance sheet **32** may not be supported by the first bar **319**. The radiation resistance sheet **32** may be supported by the first bar. When there are at least two radiation resistance sheets **32**, one radiation resistance sheet **32** may be supported on the first bar, and the other radiation resistance sheet **32** may not be supported on the first bar.

[0115] According to the above configuration, it is possible to prevent the contact between the deformed third straight portion **223** and the radiation resistance sheet **32** after the left side of the vacuum adiabatic body component assembly process. The third straight portion **223** may form an angle A with respect to the second plate **20**. The angle may be an obtuse angle. When the vacuum pressure is applied, the angle may be contracted and deformed. The contact between the radiation resistance sheet **32** and the side plate **15** may be prevented even when the angle between the radiation resistance sheet **32** and the side plate **15** is contracted and deformed. It is possible to prevent deterioration of the adiabatic performance due to the radiation resistance sheet **32**.

[0116] FIG. **15** is a side view showing the angle between the first plate and the side plate.

[0117] The angle A may be provided as an obtuse angle. Before performing the vacuum adiabatic body component assembly process, the intervening angle may be an obtuse angle. After the vacuum adiabatic body component assembly process is performed, the intervening angle may be an obtuse angle. After the vacuum adiabatic body component assembling process is performed, the intervening angle may be an acute angle.

[0118] FIG. **17** is a side view showing the angle between the first plate and the side plate.

[0119] Referring to FIG. **17**, after the vacuum adiabatic body component assembling process is performed, the intervening angle may be an acute angle B. The change in the angle may be caused by the formation of the vacuum space of the vacuum adiabatic body.

[0120] According to the embodiment, the vacuum adiabatic body that is capable of being applied to real life may be provided.

[0121] Any reference in this specification to “one embodiment,” “an embodiment,” “example

embodiment,” etc., means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of such phrases in various places in the specification are not necessarily all referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with any embodiment, it is submitted that it is within the purview of one skilled in the art to effect such feature, structure, or characteristic in connection with other ones of the embodiments.

[0122] Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

Claims

1. A vacuum adiabatic body comprising: a first plate including one surface providing a wall defining a vacuum space and the other surface; a second plate including one surface providing a wall defining the vacuum space and the other surface, the vacuum space having a thickness in a first direction between the first plate and the second plate; a side plate connecting the first plate to the second plate; and a support configured to maintain the vacuum space, wherein the second plate includes a second straight portion, and wherein the side plate includes a first straight portion provided spaced apart from the second straight portion, a third straight portion provided between the first straight portion and the second straight portion, a first curved portion provided between the first straight portion and the third straight portion, and a second curved portion provided between the third straight portion and the second straight portion.
2. The vacuum adiabatic body according to claim 1, wherein the support includes at least one of bars, a connection plate, a support plate, a porous material, or a filler.
3. The vacuum adiabatic body according to claim 1, wherein the second straight portion is provided below the first straight portion in the first direction.
4. The vacuum adiabatic body according to claim 1, wherein the side plate is deformed when a vacuum pressure is applied to the vacuum space, and wherein the side plate is bent by the vacuum pressure with the second curved portion being used as a supporting point.
5. The vacuum adiabatic body according to claim 1, wherein the first straight portion is not be in contact with the support after the vacuum adiabatic body component assembly process is performed.
6. The vacuum adiabatic body according to claim 1, wherein the third straight portion is not be in contact with the support after the vacuum adiabatic body component assembly process is performed.
7. The vacuum adiabatic body according to claim 1, wherein the first curved portion is not be in contact with the support.
8. The vacuum adiabatic body according to claim 1, wherein the second curved portion is not be in contact with the support.
9. The vacuum adiabatic body according to claim 1, wherein the support includes a porous material being not in contact with the side plate.
10. The vacuum adiabatic body according to claim 1, wherein the support includes a filler being not in contact with the side plate.
11. The vacuum adiabatic body according to claim 1, wherein the support includes bars, and the

bars are not in contact with the side plate.

12. A vacuum adiabatic body comprising: a first plate including one surface providing a wall defining a vacuum space and the other surface; a second plate including one surface providing a wall defining the vacuum space and the other surface, the vacuum space having a thickness in a first direction; a side plate connecting the first plate to the second plate; and a support configured to maintain the vacuum space, wherein the support includes a porous material being not in contact with the side plate.

13. The vacuum adiabatic body according to claim 12, wherein an empty space is provided inside the porous material.

14. The vacuum adiabatic body according to claim 12, wherein an empty space is provided between a first porous material and a second porous material.

15. The vacuum adiabatic body according to claim 12, wherein an empty space is provided between the porous material and a separate component that is distinguished from the porous material.

16. A vacuum adiabatic body comprising: a first plate including one surface providing a wall defining a vacuum space and the other surface; a second plate including one surface providing a wall defining the vacuum space and the other surface, the vacuum space having a thickness in a first direction; a side plate connecting the first plate to the second plate; and a support configured to maintain the vacuum space, wherein the support includes a filler being not in contact with the side plate.

17. The vacuum adiabatic body according to claim 16, further comprising a film provided between the first plate and the filler.

18. An appliance comprising: a vacuum adiabatic body according to claim 1.

19. An appliance comprising: a vacuum adiabatic body according to claim 12.

20. An appliance comprising: a vacuum adiabatic body according to claim 16.
